Oracle access to the model: We can query the model for any state-action pair.

* If we have a data set of N sample transitions (state, action, next state)

$$\{(s_i, a_i, s_i')\}_{i=1}^{N} \quad \text{where} \quad s_i' \sim P(\cdot \mid s_i, a_i) \quad \forall i \in \{1, 2, \dots, N\},$$

we can form an estimate transition function $\hat{P}$.

Estimating the transition function: We can sample n transitions for each state-action pair and then compute $\hat{P}$.

1) Data collection : For all $s \in \mathcal{S}$ and $a \in \mathcal{A}$, sample $s' \sim P(\cdot \mid s, a)$ n times $\longrightarrow N = |\mathcal{S}||\mathcal{A}|\, n$ samples

2) Model estimation : For all $s \in \mathcal{S}$ and $a \in \mathcal{A}$, compute

$$\hat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbb{1}[s_i = s, a_i = a, s_i' = s']}{\underbrace{\sum_{i=1}^{N} \mathbb{1}[s_i = s, a_i = a]}_{n}}$$

For each $s \in \mathcal{S}$, $a \in \mathcal{A}$:

$$P(\cdot \mid s, a) \longrightarrow \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{|\mathcal{S}|} \end{bmatrix}$$

$$\forall j : p_j \geq 0 \qquad \sum_{j=1}^{|\mathcal{S}|} p_j = 1$$

Example:

Data set

state s	action a	next state s'
$\vdots$	$\vdots$	$\vdots$
2	1	2
2	1	4
2	1	4
$\vdots$	$\vdots$	$\vdots$
2	1	7
$\vdots$	$\vdots$	$\vdots$

n samples for $s=2, a=1$

Consider $s=2$, $a=1$ $\longrightarrow$

$$\begin{cases} P(s'=1 \mid s=2, a=1) = \dfrac{\#(2,1,1)}{n} \\ \quad\vdots \\ P(s'=4 \mid s=2, a=1) = \dfrac{\#(2,1,4)}{n} \\ \quad\vdots \\ P(s'=|S| \mid s=2, a=1) = \dfrac{\#(2,1,|S|)}{n} \end{cases}$$

Estimation error for the transition function: With finite samples of transitions (s, a, s'), one cannot learn the transition function perfectly in general.

Question: How can we bound the estimation error of the transition function with finite samples?

Using concentration inequalities

Bounded Difference Property: A function $f: \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n \longrightarrow \mathbb{R}$ satisfies the bounded difference property if changing any one of its arguments results in a bounded change in its output. Formally, there should exist constants $c_1, c_2, \dots, c_n$ such that for all $i \in \{1, 2, \dots, n\}$ and for all $x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, \dots, x_n \in \mathcal{X}_n, x_i' \in \mathcal{X}_i$,

$$\max_{\substack{x_1, x_2, \dots, x_n \\ x_i'}} \left| f(x_1, x_2, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) - f(x_1, x_2, \dots, x_{i-1}, x_i', x_{i+1}, \dots, x_n) \right| \leqslant c_i .$$

McDiarmid's Inequality: Consider independent random variables $X_1 \in \mathcal{X}_1, X_2 \in \mathcal{X}_2, \dots, X_n \in \mathcal{X}_n$. Let $f: \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n \longrightarrow \mathbb{R}$ be a function satisfying the bounded difference property with bounds $c_1, c_2, \dots, c_n$, corresponding to the n arguments of f.

For any $\varepsilon > 0$, the following holds:

$$\mathbb{P}\left[\left| f(X_1, X_2, \dots, X_n) - \mathbb{E}\left[f(X_1, X_2, \dots, X_n)\right] \right| \geqslant \varepsilon \right] \leqslant 2\exp\left(\frac{-2\varepsilon^2}{\sum_{i=1}^{n} c_i^2} \right).$$

Lemma [Estimation of Categorical Distribution]: Let X be a discrete random variable over $\mathcal{X} = \{1, 2, \dots, d\}$ with categorical distribution

$$\vec{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_d \end{bmatrix} \quad \text{where } q_i \geq 0 \ \forall j \in \{1, 2, \dots, d\} \text{ and } \sum_{j=1}^{d} q_j = 1.$$

Assume we have n i.i.d. samples $x_1, x_2, \dots, x_n$ of X and form an empirical estimate of $\vec{q}$ as

$$\vec{\hat{q}} = \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \vdots \\ \hat{q}_d \end{bmatrix}, \text{ where } \hat{q}_j = \frac{\sum_{i=1}^{n} \mathbb{1}[x_i = j]}{n}.$$

$\mathbb{P}(\text{error} \geq \varepsilon') \leq \delta$ — low error with

$\mathbb{P}(\text{error} < \varepsilon') \geq 1 - \delta$ — high probability

Then, for any $\varepsilon > 0$, it holds that

$$\boxed{\mathbb{P}\left(\|\vec{\hat{q}} - \vec{q}\|_2 \geq \frac{1}{\sqrt{n}} + \varepsilon\right) \leq e^{-n\varepsilon^2}.}$$

Proof: Let $\tilde{e}_i$ be a random variable (vector) defined based on x_i as

$$\tilde{e}_i = e_j \text{ if } x_i = j, \quad \text{where } e_j = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{d \times 1} \begin{matrix} \\ \\ \\ \rightarrow j^{th} \text{ entry} \\ \\ \\ \\ \end{matrix}.$$

Then, $\vec{\hat{q}} = \dfrac{\sum_{i=1}^{n} \tilde{e}_i}{n}$.

Consider $\vec{\hat{q}}' = \dfrac{\sum_{i=1}^{n} \tilde{e}_i'}{n}$, where $\tilde{e}_i' = \tilde{e}_i$ for all $i \in \{1, 2, \dots, n\}$ except one.

$$\boxed{\begin{matrix} f(x_1, & x_2, & \dots, & x_n) = \|\vec{\hat{q}} - \vec{q}\|_2 \\ \downarrow & \downarrow & & \downarrow \\ \tilde{e}_1 & \tilde{e}_2 & & \tilde{e}_n \end{matrix}}$$

Then, we have

at most two entries of $\vec{\hat{q}}$ & $\vec{\hat{q}}'$ are different by $\frac{1}{n}$

$$\underbrace{\|\vec{\hat{q}}-\vec{q}\|_2}_{f(\tilde{e}_1,\tilde{e}_2,\dots,\tilde{e}_n)} - \underbrace{\|\vec{\hat{q}}'-\vec{q}\|_2}_{f(\tilde{e}_1,\dots,\tilde{e}_{i-1},\tilde{e}_i',\tilde{e}_{i+1},\dots,\tilde{e}_n)} \overset{\text{triangle inequality}}{\leq} \overbrace{\|\vec{\hat{q}}-\vec{\hat{q}}'\|_2}^{\leq\sqrt{(\frac{1}{n})^2+(\frac{1}{n})^2}} \leq \underbrace{\frac{\sqrt{2}}{n}}_{c_i}.$$

$$\boxed{f(\tilde{e}_1,\tilde{e}_2,\dots,\tilde{e}_n)=\|\vec{\hat{q}}-\vec{q}\|_2}$$

Therefore, applying McDiarmid's inequality yields

$$\mathbb{P}\big(\underbrace{\|\vec{\hat{q}}-\vec{q}\|_2}_{f(\cdot)} - \underbrace{\mathbb{E}[\|\vec{\hat{q}}-\vec{q}\|_2]}_{\mathbb{E}[f(\cdot)]} \geq \varepsilon\big) \overset{\text{one-sided version}}{\leq} \exp\Big(\frac{-2\varepsilon^2}{\sum_{i=1}^n c_i^2}\Big)$$

$$= \exp\Big(\frac{-2\varepsilon^2}{n\times\frac{2}{n^2}}\Big) = e^{-n\varepsilon^2}$$

Now, we need to evaluate $\mathbb{E}[\|\vec{\hat{q}}-\vec{q}\|_2]$.

$$\mathbb{E}[\|\vec{\hat{q}}-\vec{q}\|_2] \overset{\text{definition of } \vec{\hat{q}}}{=} \mathbb{E}\Big[\Big\|\frac{\sum_{i=1}^n \tilde{e}_i}{n}-\vec{q}\Big\|_2\Big]$$

$g^{-1}(g(x))=x$

$$= \mathbb{E}\Big[\sqrt{\Big\|\frac{\sum_{i=1}^n \tilde{e}_i}{n}-\vec{q}\Big\|_2^2}\Big]$$

Jensen's inequality

$$\leq \sqrt{\mathbb{E}\Big[\Big\|\frac{\sum_{i=1}^n \tilde{e}_i}{n}-\vec{q}\Big\|_2^2\Big]}$$

$$= \frac{1}{n^2}\mathbb{E}\Big[\Big\|\sum_{i=1}^n \tilde{e}_i - n\vec{q}\Big\|_2^2\Big] = \frac{1}{n^2}\mathbb{E}\Big[\Big\|\sum_{i=1}^n (\tilde{e}_i-\vec{q})\Big\|_2^2\Big]$$

triangle inequality + Linearity of expectation

$$\leq \sqrt{\frac{1}{n^2}\sum_{i=1}^n \mathbb{E}\big[\underbrace{\|\tilde{e}_i-\vec{q}\|_2^2}_{(\tilde{e}_i-\vec{q})^T(\tilde{e}_i-\vec{q})}\big]}$$

definition of L2 norm

$$= \sqrt{\frac{1}{n^2}\sum_{i=1}^n \mathbb{E}\big[\tilde{e}_i^T\tilde{e}_i - 2\tilde{e}_i^T\vec{q} + \vec{q}^T\vec{q}\big]}$$

Linearity of expectation

$$= \sqrt{\frac{1}{n^2}\sum_{i=1}^n \Big(\underbrace{\mathbb{E}[\tilde{e}_i^T\tilde{e}_i]}_{1} - 2\underbrace{\mathbb{E}[\tilde{e}_i^T\vec{q}]}_{\mathbb{E}[\tilde{e}_i^T]\vec{q}=\vec{q}^T\vec{q}=\|\vec{q}\|_2^2} + \underbrace{\mathbb{E}[\|\vec{q}\|_2^2]}_{\|\vec{q}\|_2^2}\Big)}$$

$(\tilde{e}_i)_{i=1}^n$ are i.i.d.

$$= \sqrt{\frac{1}{n^2}\, n\,(1-\|\vec{q}\|_2^2)}$$

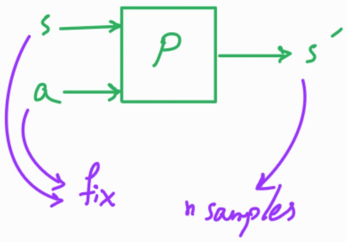
$$= \sqrt{\frac{1}{n}(1-\|\vec{q}\|_2^2)} \leq \frac{1}{\sqrt{n}}$$

Finally, we obtain the desired result:

$$\mathbb{P}\left(\| \hat{\vec{q}} - \vec{q} \|_2 \geqslant \frac{1}{\sqrt{n}} + \varepsilon \right) \leqslant e^{-n\varepsilon^2}.$$

$\blacksquare$

Theorem [Estimation of Transition Function]: Suppose we have access to an oracle model of the transition function through which we collect N sample transitions to form the estimate



$$\hat{P}(s'|s,a) = \frac{\sum_{i=1}^{N} \mathbb{1}[s_i = s, a_i = a, s_i' = s']}{\sum_{i=1}^{N} \mathbb{1}[s_i = s, a_i = a]}$$

Then, with probability at least $1-\delta$ for any $\delta > 0$, the following holds:

$$\boxed{\max_{\substack{s\in S \\ a \in A}} \| \hat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 \leq \sqrt{\frac{|S|^2 |A| \log\left(\frac{2|S||A|}{\delta}\right)}{N}}}$$

Proof (sketch): The proof relies on the following arguments:

- for a single (s,a) pair ← Lemma about estimation of categorical distribution
- $\|\cdot\|_2 \to \|\cdot\|_1$ ← Equivalence of norms $\|E\|_2 \leq \|E\|_1 \leq \sqrt{d}\,\|E\|_2$ for $E \in \mathbb{R}^d$
- for all (s,a) pairs ← Union bound $\mathbb{P}[B_1 \cup B_2 \cup B_3 \cup \ldots \cup B_p] \leq \sum_{j=1}^{p} \underbrace{\mathbb{P}[B_j]}_{\leq \frac{\varepsilon}{|S||A|}}$ $\Rightarrow \mathbb{P}\left[\bigcup_{j=1}^{p} B_j\right] \leq \varepsilon$

  $p = |S||A|$

Question: *How does the suboptimality of the learned policy depend on the quality of estimating the transition function?*

Occupancy Measures:

$$\boxed{\sum_{s \in S} \rho_{s_0}^{\pi}(s) = \frac{1}{1-\gamma}}$$

state occupancy measure $\rho_{s_0}^{\pi}(s) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s \mid s_0, \pi, P)$

normalized state occupancy measure $\bar{\rho}_{s_0}^{\pi}(s) = (1-\gamma)\, \rho_{s_0}^{\pi}(s)$

$$\boxed{\sum_{s \in S} \bar{\rho}_{s_0}^{\pi}(s) = 1}$$

state-action occupancy measure $\nu_{s_0}^{\pi}(s,a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s, A_t = a \mid s_0, \pi, P)$

normalized state-action occupancy measure $\bar{\nu}_{s_0}^{\pi}(s,a) = (1-\gamma)\, \nu_{s_0}^{\pi}(s,a)$

Simulation Lemma: Consider a deterministic policy $\pi: S \to A$ and two transition functions P and $\hat{P}$ over an infinite-horizon discounted MDP. The value function of policy π under P and $\hat{P}$ satisfies the following:

$$\boxed{\left| \hat{V}^{\pi}(s_0) - V^{\pi}(s_0) \right| \leq \frac{\gamma R_{max}}{(1-\gamma)^2} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi}} \Big[\underbrace{\left\| \hat{P}(\cdot \mid s, \pi(s)) - P(\cdot \mid s, \pi(s)) \right\|_1}_{} \Big]},$$

where $R_{max} = \max_{s,a} |R(s,a)|$.

capturing $TVD(\hat{P}(\cdot \mid s, \pi(s)), P(\cdot \mid s, \pi(s)))$

total variation distance

> Intuitively:
> - Not frequently visited states do not matter
> - Actions not taken by the policy do not matter
> - If $\hat{P} \approx P \longrightarrow \hat{V} \approx V$

Proof: Consider $\hat{V}^\pi$ and V^π as vector representation of the value function of π under $\hat{P}$ and P, respectively.

$$\hat{V}^\pi - V^\pi \overset{\text{Bellman consistency equation}}{=} R^\pi + \gamma \hat{P}^\pi \hat{V}^\pi - R^\pi - \gamma P^\pi V^\pi$$

$$
\begin{aligned}
\text{rearrange terms} \leftarrow \quad &= \gamma \left(\hat{P}^\pi \hat{V}^\pi - P^\pi V^\pi \right) \\
\text{add and subtract} \leftarrow \quad &= \gamma \left(\hat{P}^\pi \hat{V}^\pi \underline{- P^\pi \hat{V}^\pi} + \underline{P^\pi \hat{V}^\pi} - P^\pi V^\pi \right) \\
\text{rearrange terms} \leftarrow \quad &= \gamma \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi + \gamma P^\pi \left(\underline{\underline{\hat{V}^\pi - V^\pi}} \right) \\
\text{expand } \hat{V}^\pi - V^\pi \leftarrow \quad &= \gamma \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi + \gamma P^\pi \left(\underline{\underline{\gamma \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi + \gamma P^\pi \left(\hat{V}^\pi - V^\pi \right)}} \right) \\
\text{rearrange terms} \leftarrow \quad &= \gamma \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi + \gamma^2 P^\pi \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi + \gamma^2 \left(P^\pi \right)^2 \left(\hat{V}^\pi - V^\pi \right) \\
&\;\vdots \\
\text{after } K \text{ iterations} \leftarrow \quad &= \sum_{i=1}^{K} \left(\gamma^i \left(P^\pi \right)^{i-1} \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi \right) + \underbrace{\gamma^K \left(P^\pi \right)^K \left(\hat{V}^\pi - V^\pi \right)}_{\text{goes to zero as } K \to \infty} \\
&\;\vdots \\
K \to \infty \leftarrow \quad &= \sum_{i=1}^{\infty} \left(\gamma^i \left(P^\pi \right)^{i-1} \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi \right)
\end{aligned}
$$

Therefore, we have that

$$\hat{V}^\pi - V^\pi = \gamma \sum_{i=0}^{\infty} \gamma^i \left(P^\pi \right)^i \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi .$$

Now, we compare the value function at the initial state.

$$\hat{V}^\pi(s_0) - V^\pi(s_0) = \left(\hat{V}^\pi - V^\pi \right)^T e_{s_0} \longrightarrow \text{standard basis vector} \quad \begin{array}{cl} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{|S| \times 1} & \begin{array}{l} \\ \\ \\ \rightarrow s_0^{\text{th}} \text{ entry} \\ \\ \\ \\ \end{array} \end{array}$$

$$\text{based on above} \leftarrow \quad = \left(\gamma \sum_{i=0}^{\infty} \gamma^i \left(P^\pi \right)^i \left(\hat{P}^\pi - P^\pi \right) \hat{V}^\pi \right)^T e_{s_0}$$

rearrange terms

$$= \gamma \left((\hat{P}^\pi - P^\pi)\, \hat{V}^\pi \right)^T \sum_{i=0}^{\infty} \gamma^i \left((P^\pi)^i\right)^T e_{s_0}$$

definition of $\bar{\rho}^\pi_{s_0}$

$$= \frac{\gamma}{1-\gamma} \left((\hat{P}^\pi - P^\pi)\, \hat{V}^\pi \right)^T \bar{\rho}^\pi_{s_0}$$

definiton of inner product

$$= \frac{\gamma}{1-\gamma} \sum_{s \in \mathcal{S}} \left((\hat{P}^\pi - P^\pi)\, \hat{V}^\pi \right)_s \bar{\rho}^\pi_{s_0}(s)$$

s th entry

definition of expectation

$$= \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\left((\hat{P}^\pi - P^\pi)\, \hat{V}^\pi \right)_s \right]$$

definition of matrix multiplication

$$= \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[(\hat{P}^\pi - P^\pi)_s\, \hat{V}^\pi \right]$$

s th row

definiton of inner product

$$= \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\sum_{s' \in \mathcal{S}} \left(\hat{P}^\pi(s' \mid s, \pi(s)) - P^\pi(s' \mid s, \pi(s)) \right) \hat{V}^\pi(s') \right]$$

Taking absolute value, we have

$$\left| \hat{V}^\pi(s_0) - V^\pi(s_0) \right| = \left| \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\sum_{s' \in \mathcal{S}} \left(\hat{P}^\pi(s' \mid s, \pi(s)) - P^\pi(s' \mid s, \pi(s)) \right) \hat{V}^\pi(s') \right] \right|$$

triangle inequality

$$\leq \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\sum_{s' \in \mathcal{S}} \left| \hat{P}^\pi(s' \mid s, \pi(s)) - P^\pi(s' \mid s, \pi(s)) \right| \left| \hat{V}^\pi(s') \right| \right]$$

bound on value function

$$\leq \frac{\gamma}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\sum_{s' \in \mathcal{S}} \left| \hat{P}^\pi(s' \mid s, \pi(s)) - P^\pi(s' \mid s, \pi(s)) \right| \frac{R_{max}}{1-\gamma} \right]$$

definition of L1 norm

$$= \frac{\gamma R_{max}}{(1-\gamma)^2} \mathbb{E}_{s \sim \bar{\rho}^\pi_{s_0}} \left[\left\| \hat{P}(\cdot \mid s, \pi(s)) - P(\cdot \mid s, \pi(s)) \right\|_1 \right].$$

$\blacksquare$

Corollary: In the setting considered in the simulation Lemma, it holds that

$$\left| \hat{V}^\pi(s_0) - V^\pi(s_0) \right| \leq \frac{\gamma R_{max}}{(1-\gamma)^2} \overbrace{\max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} \left\| \hat{P}(\cdot \mid s, a) - P(\cdot \mid s, a) \right\|_1}^{\text{no dependence on policy}}.$$

Sub-optimality due to model inaccuracy:

The error in estimating the model, the transition function P and reward function R, leads to sub-optimality of the policy learned from them.

Note:

- Assume that R is known and $\hat{P}$ is an estimate of P.
- Let $\hat{V}^{\pi}$ be the value function of a policy π evaluated using $\hat{P}$.
- Let $\hat{\pi}^*$ be the policy learned using $\hat{P}$ as an optimal policy.

The sub-optimality of $\hat{\pi}^*$ is:

for a π^*

$$0 \leq V^*(s_0) - V^{\hat{\pi}^*}(s_0) = V^{\pi^*}(s_0) - V^{\hat{\pi}^*}(s_0)$$

add & subtract term

$$= V^{\pi^*}(s_0) - \hat{V}^{\hat{\pi}^*}(s_0) + \hat{V}^{\hat{\pi}^*}(s_0) - V^{\hat{\pi}^*}(s_0)$$

$\hat{V}^{\pi^*} \leq \hat{V}^{\hat{\pi}^*}$

$$\leq \underbrace{V^{\pi^*}(s_0) - \hat{V}^{\pi^*}(s_0)}_{\pi^* \text{ evaluated over two models}} + \underbrace{\hat{V}^{\hat{\pi}^*}(s_0) - V^{\hat{\pi}^*}(s_0)}_{\hat{\pi}^* \text{ evaluated over two models}}$$

$\longrightarrow$

$$\left| V^*(s_0) - V^{\hat{\pi}^*}(s_0) \right| \leq \left| V^{\pi^*}(s_0) - \hat{V}^{\pi^*}(s_0) + \hat{V}^{\hat{\pi}^*}(s_0) - V^{\hat{\pi}^*}(s_0) \right|$$

$$\leq \left| V^{\pi^*}(s_0) - \hat{V}^{\pi^*}(s_0) \right| + \left| \hat{V}^{\hat{\pi}^*}(s_0) - V^{\hat{\pi}^*}(s_0) \right|$$

applying the corollary from the simulation lemma

$$\leq \frac{2\gamma R_{max}}{(1-\gamma)^2} \max_{\substack{s \in S \\ a \in A}} \left\| \hat{P}(\cdot|s,a) - P(\cdot|s,a) \right\|_1$$

Theorem [sub-optimality of policy]: In the setting with oracle access to the model, for any $\delta > 0$ and $\epsilon > 0$, the model-based RL algorithm with N samples

$$N \geq \frac{4\gamma^2 R_{max}^2 |S|^2 |A| \log\left(\frac{2|S||A|}{\delta}\right)}{\epsilon^2 (1-\gamma)^4}$$

will result in $\hat{\pi}^*$ such that with probability at least $1-\delta$,

$$V^*(s) - V^{\hat{\pi}^*}(s) \leq \epsilon \qquad \forall s \in S.$$